

Supplementary Material for *When Can Quantum Event Classifiers Be Trusted? Conditional Validity under Collider Systematics and Quantum Estimation Uncertainty*

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This supplement gives the full algebra behind the formal results, the registered-falsifier and correction trail, and extended numerical tables. Every numerical table is a transcription or an explicitly identified summary of the immutable JSON artifact named in its caption. Superseded first-run tables remain in the repository under `*_v1_*.json`.

1 Formal details

1.1 Exact weighted anytime-valid certification

For IID labeled draws (c_i, u_i) , with $c_i \in \{0, 1\}$, $u_i \in [0, w_{\max}]$, $E[u] > 0$, and $R = E[uc]/E[u]$, fix a claim $R \geq \tau$ and define

$$Z_i(\tau) = \frac{u_i(c_i - \tau) + \tau w_{\max}}{w_{\max}}.$$

Because $u_i(c_i - \tau)$ lies between $-\tau w_{\max}$ and $(1 - \tau)w_{\max}$, $Z_i(\tau) \in [0, 1]$. Moreover,

$$E[Z(\tau)] \geq \tau \iff E[u(c - \tau)] \geq 0 \iff E[uc] \geq \tau E[u] \iff R \geq \tau.$$

The reduction is therefore exact. If (L_n, U_n) is any level $1 - \alpha$ confidence sequence for the mean of Z , then on its time-uniform coverage event $L_n \geq \tau$ implies $R \geq \tau$ at every n . Thus

$$P\{\exists n : \text{SUPPORTED and } R < \tau\} \leq \alpha$$

under arbitrary stopping. Setting $u \equiv 1$ and $w_{\max} = 1$ recovers the unweighted construction. Weighting changes the variance and hence the stopping time, but adds no validity slack.

1.2 Information-set boundary for weight-only nuisances

For a weight-only nuisance $P_\theta(X) = P_0(X)$ and an unchanged correctness process. Hence every I1 statistic has the same sampling law under θ and 0, and so does I2 when labels carry nominal weights. Any size- α test based on those observations has power exactly α against the nuisance. A fail-closed auditor must leave any claim whose truth changes only through the hidden rate weights UNRESOLVED.

At I3 a control-region count $N \sim \text{Poisson}(\lambda(\theta))$ with $\lambda(\theta) \neq \lambda(0)$ changes the sampling law and restores non-trivial power. The restoration is quantitative rather than absolute: template variance

$$\sigma_c^2 = \sum_g (\text{relerr}_g \lambda_g)^2$$

sets the resolution scale. This is why the template-statistics-aware E14 fit identifies the ttbar rate only to approximately $\pm 10\%$ and leaves tighter claims unresolved.

1.3 Random deployments and verdict stability

Let ω denote the finite-shot or hardware realization and $\tilde{f}_\omega$ the resulting refit, recalibration, and threshold. If the audit controls false certification conditionally,

$$P(\text{false certification} \mid \omega) \leq \alpha,$$

then the tower property gives marginal control for both the deployment-relative and ideal-anchored claim classes:

$$P(\text{false certification}) = E_\omega[P(\text{false certification} \mid \omega)] \leq \alpha.$$

For signed target and source movements ΔM_T and ΔM_S , the ideal-anchored margin moves by at most $|\Delta M_T|$, while the deployment-relative margin moves by $|\Delta M_T - \Delta M_S|$. On the confidence-sequence coverage event, if the movement is smaller than the ideal margin and the realized audit resolves, the verdict sign is unchanged. For weighted claims the metric-scale margin is mapped to the certification scale by $E[u]/w_{\max}$. Common-mode movement cancels from the deployment-relative margin but not from the ideal-anchored margin. Figure S16 and Table S6 show the measured movement distributions that instantiate these conditions; no first-principles Frobenius-to-metric bound is claimed.

1.4 Pre-split component allocation for weighted balanced accuracy

For $\text{BA}_w = (\text{TPR}_w + \text{TNR}_w)/2 \geq \tau_{\text{BA}}$, predeclare component thresholds τ_1, τ_2 satisfying $(\tau_1 + \tau_2)/2 = \tau_{\text{BA}}$ and error levels $\alpha_1 + \alpha_2 = \alpha$. Each component uses the exact one-sample reduction with its predeclared class-specific bound. The BA claim is SUPPORTED only if both components are supported, REFUTED only if both are refuted, and UNRESOLVED otherwise. If BA is false, at least one component claim is false; certifying it requires a coverage violation. A union bound therefore controls false certification at α . The symmetric argument controls false refutation. Table S9 shows that the removable ratio-CS slack disappears on the synthetic control, while the benchmark’s weighted signal fraction makes the physics case information-limited.

2 Registered falsifiers and audit trail

The count below is by *unique registered falsifier arm*. The E15 calibration gate is one arm, although it rejected two successive invalid implementations before the valid grid was allowed to run. This convention reconciles the nine-arm tally with the two gate interventions.

Table S1: All nine registered falsifier arms that fired. Frozen conditions and consequences are recorded in `docs/experiment_registry.md`.

Arm	Frozen condition	Observed firing	Consequence obeyed
E02R	TES sign pattern fails multi-seed replication	Up-shift monotonicity in only 2/5 seeds	Symmetric TES narrative withdrawn; only small down-shift degradation retained.
E12(e)	Two flagship decoupling cells fail to reproduce	Diboson flagship cells stayed nominal-like in the fresh world	Normalization-collapse statement scoped to signal-region composition; failing registered cells kept.
E14 v1	Control-region interval coverage below nominal	Pure-Poisson fit gave ttbar bias +0.07 and zero coverage	Run blocked; template-statistics likelihood registered, then rerun as v2.
E15 gate	Nominal L2 coverage outside 0.6827 ± 0.02	Conditional-ensemble and numerical-conditioning implementations each failed	Both grids blocked; ensemble semantics and optimizer fixed before any shifted claim.
E17(b)	Degradation signs fail across additional worlds	One world improved QK under combo by 0.009; another under TES-down by 0.005	Cross-world degradation claim withdrawn.
E08v2(a)	Corrected counting coverage outside the nominal band	One-draw corrected coverage 0.9916–1.000	Arm blocked and bounded multi-draw E08v3 registered.
E08v2(b)	Independent-template flagship L2 coverage < 0.633	Coverage 0.000 versus shared 0.7188	“Profiling restores validity” made explicitly shared-simulation-conditional.
E08v3(a)	Marginal symmetric-BB coverage outside 0.6827 ± 0.06 for at least two gated models	All three gated models overcovered, 0.7513–0.8284	Calibration-restoration claim withdrawn; conservative direction reported.
E13v2(b)	All true benchmark BA claims unresolved by 5,000 labels	TPR component 0/200 resolutions in every cell	BA path published as information-limited; TNR success and mechanism reported separately.

Table S2: Audit-trail summary. The full findings and dispositions are in `docs/audits/`.

Audit	Scope	Material effect
Pre-campaign, 2026-08-11	All E00–E11 code, tables, splits, and governance	Seven design constraints frozen before E12–E16: archived row indices, role separation, post-selection sensor validation, an independent floor, independent quantum-noise streams, a typed deployment snapshot, and weighted estimands. No prior headline invalidated.
Post-campaign, 2026-08-11	E12–E16, E04v3, E11v2; 172 registry values and about 178 manuscript claims	E13 nominal-weight estimand fixed and rerun; E15 gate rejected two implementations; E16 dual accounting added; E12 normalization reading corrected; all superseded tables retained.
Pre-submission, 2026-08-12	Extension runners, figure scripts, and line-by-line formal proofs	Two HIGH findings fixed: JES tracking scoped and disclosed; E19 weighted estimand corrected and rerun. Proposition 4 gained its coverage-event and scale-bridge qualifiers. Eight LOW findings were fixed, completed, disclosed, or documented.
F8.2, 2026-08-12	Final LaTeX, supplement, number trace, semantic equivalence, rendered PDFs	E08v3 bias range corrected from +6.1 to the archived +7.5 endpoint; the nine-arm counting convention made explicit.

3 Extended numerical results

3.1 Inference sensitivity (E15)

Table S3 reports the local slope at the nominal point, averaged over the five injected μ values. L1 is the counting estimator, L2 the complete profile, and L3 the profile with the shifted family omitted. Only the three models that passed the nominal L2 gate are shown. Units are μ per nuisance unit: per standard deviation for TES and JES, per GeV for soft-MET, and per unit scale for the normalization families.

Table S3: E15 local sensitivity and nuisance tracking. Source: `results/tables/E15_sensitivity.json`.

Family	Model	L1 slope	L2 slope	L3 slope	L2 tracking
TES	QK-SVC	9.8095	0.4092	7.1213	0.9970
TES	RBF-SVC	10.3065	0.2269	8.6641	0.9986
TES	XGBoost	5.4959	0.1031	4.7948	0.9938
JES	QK-SVC	3.6530	-0.2033	5.1238	0.9813
JES	RBF-SVC	2.7528	-0.1096	5.7069	0.9901
JES	XGBoost	1.7328	0.0121	1.1957	0.7116
soft-MET	QK-SVC	-0.0812	-0.2126	6.5142	0.2497
soft-MET	RBF-SVC	0.7731	-0.0496	3.0473	0.2511
soft-MET	XGBoost	0.7305	-0.4605	-1.5627	0.5001
ttbar scale	QK-SVC	49.7510	0.5175	-0.7125	1.0040
ttbar scale	RBF-SVC	54.0570	-1.1795	-4.0830	0.9915
ttbar scale	XGBoost	12.7935	-0.0220	9.0305	0.9895
diboson scale	QK-SVC	4.7626	0.0834	0.5676	0.9900
diboson scale	RBF-SVC	4.3356	0.2769	1.0303	0.9869
diboson scale	XGBoost	0.9981	-0.0065	0.4824	1.0006
background scale	QK-SVC	270.1200	-30.9900	222.4700	1.0000
background scale	RBF-SVC	397.2600	34.5600	823.4100	0.9800
background scale	XGBoost	101.8800	-5.3900	-3.6800	1.0000

3.2 Fresh-world validity and stopping-time efficiency

Table S4: Fresh-world validity replication. Source: `results/tables/E19_fresh_world_validity.json`; the false-refutation counts documented in the registered status agree with the rates stored in the table.

Accounting	False certification	False refutation	$\alpha + 3\sigma$ boundary
Unweighted, non-vetoed	11/3,060 = 0.359%	1/11,980 = 0.008%	6.182%
Unweighted, all streams	11/7,700 = 0.143%	1/11,980 = 0.008%	6.182%
Weighted, nominal weights	6/7,980 = 0.075%	4/11,700 = 0.034%	5.732%

All 12 archived model–environment score comparisons were byte-identical before auditing. The weighted bound was $w_{\max} = 7.22421 \times 2.05 = 14.80963$. The corrected nominal-weight estimand is invariant across all 12 weight-only environments.

Table S5: Stopping-time price relative to the Wald information yardstick. Source: `results/tables/E06_nstar_efficiency.json`.

Subset	Cells	Q25	Median	Q75
All resolved cells	518	1.56	2.07	2.97
Margin $[0.01, 0.02)$	95	2.82	3.35	3.85
Margin $[0.02, 0.04)$	95	2.62	3.19	3.66
Margin $[0.04, 0.08)$	164	1.63	1.98	2.38
Margin ≥ 0.08	164	1.25	1.46	1.72
Certification direction	487	1.54	1.99	2.78
Refutation direction	31	3.12	3.34	3.65

3.3 Quantum-estimation diagnostics (E16)

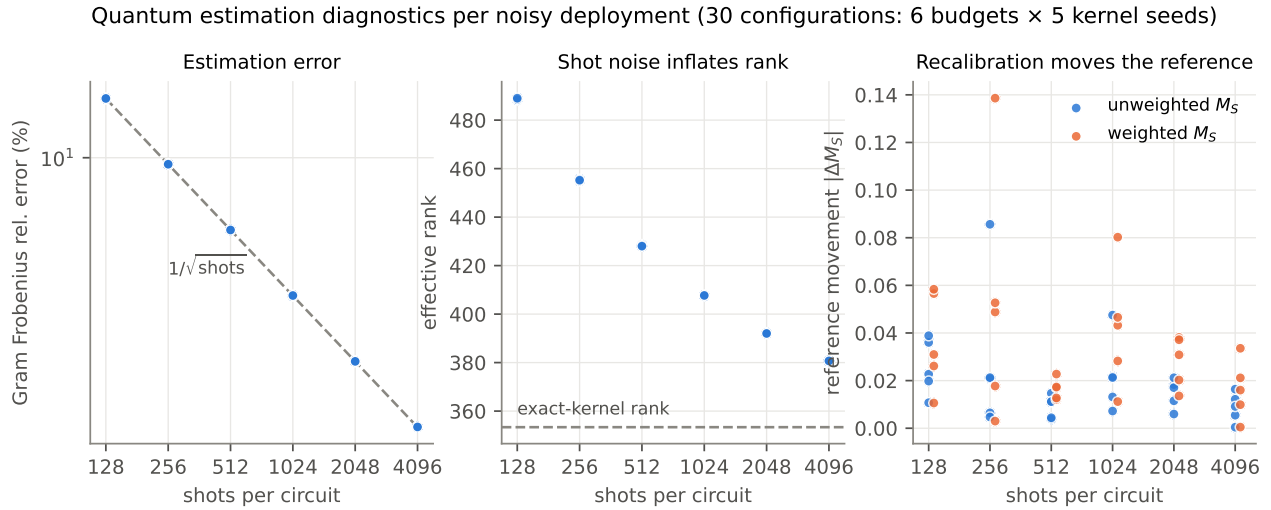


Figure S16: Per-configuration diagnostics for all 30 noisy deployments: relative Frobenius error, effective-rank inflation, and source reference movement. The empirical distributions, rather than a claimed analytic kernel-to-metric bound, calibrate the stability conditions. Source: `results/tables/E16_quantum_uncertainty.json`.

Table S6: E16 shot-budget summary. Frobenius error and effective rank are means over five deployments; movement is the maximum absolute weighted source-reference shift. Flip and abstention entries are percentages. Source: `results/tables/E16_quantum_uncertainty.json`.

Shots	Frob. err. (%)	Rank	max $ \Delta M_S $	Far flips		Moderate flips		Near abst.
				own	fixed	own	fixed	own
128	13.66	488.6	0.058	0.0	20.8	15.8	71.4	94.0
256	9.66	455.2	0.139	0.0	17.7	16.1	49.4	92.9
512	6.84	428.3	0.023	0.0	0.9	11.5	47.7	92.5
1024	4.83	407.6	0.080	0.0	11.9	13.2	54.6	92.0
2048	3.42	392.1	0.038	0.0	5.8	11.1	58.4	93.9
4096	2.42	380.7	0.034	0.0	0.4	6.7	40.4	93.0

3.4 Independent-MC split studies (E08v3)

Table S7: Marginal nominal counting coverage over 400 independent splits. Source: `results/tables/E08v3_multidraw.json`.

Model	Shared truth	Indep. naive	Indep. BB	Indep. BB symmetric	Stored prediction
QK-SVC	0.6828	0.0691	0.6501	0.8113	0.5220
RBF-SVC	0.6827	0.0860	0.6605	0.8284	0.5220
XGBoost (A)	0.6830	0.0673	0.5650	0.7513	0.5216
XGBoost (B)	0.6823	0.0645	0.5458	0.7033	0.5164

Table S8: Independent-template profile results across 10 splits, averaged over the frozen μ grid within each split. Source: `results/tables/E08v3_multidraw.json`.

Cell	Coverage range	Draws < 0.633	Bias range	Shared reference
TES = 0.98, XGBoost	0.000–0.238	10/10	−6.600–+4.050	0.7188
Nominal, XGBoost	0.000–0.359	10/10	−6.557–+7.496	gate-passing in E15

3.5 Weighted balanced-accuracy allocation (E13v2)

Table S9: E13v2 component-allocation battery, 200 replications per cell at $n_{\max} = 5,000$. Entries are correct resolutions; all false-certification counts are zero. Source: `results/tables/E13v2_baw_allocation.json`.

Population	Claim	BA pre-split	TPR component	TNR component	v1 BA	median n^*
Synthetic	$m = 0.02$, true	0/200	9/200	36/200	0/200	–
Synthetic	$m = 0.02$, false	0/200	3/200	36/200	0/200	–
Synthetic	$m = 0.05$, true	83/200	83/200	199/200	0/200	3,489
Synthetic	$m = 0.05$, false	87/200	87/200	200/200	0/200	3,370
Benchmark	$m = 0.02$, true	0/200	0/200	67/200	0/200	–
Benchmark	$m = 0.02$, false	0/200	0/200	64/200	0/200	–
Benchmark	$m = 0.05$, true	0/200	0/200	200/200	0/200	–
Benchmark	$m = 0.05$, false	0/200	0/200	200/200	0/200	–

Table S10: Mechanism diagnostics for E13v2 at $n = 5,000$.

Diagnostic	Synthetic control	Benchmark physics weights
Weighted signal fraction	0.301459	0.000966
Positive-class ESS ratio	0.226298	0.001106
$w_{\max}^{(+)}$	4.58547	5.98645
$w_{\max}^{(-)}$	9.40021	14.89144
TPR Z -margin at $m = 0.05$	0.01130253	0.00002827
TPR CS radius	0.011776	0.001806
v1 BA radius	0.168819	0.288855

In the benchmark, the TPR radius-to-margin ratio is about 64 at 5,000 labels. A quadratic information-scaling extrapolation therefore gives $n^* \approx 5,000 \times 64^2 \simeq 2 \times 10^7$ uniform labels at margin 0.05. This is a diagnostic extrapolation, not a new Monte Carlo run.

4 Artifact map

The main machine-readable sources for this supplement are:

- `results/tables/E15_sensitivity.json`;
- `results/tables/E19_fresh_world_validity.json`;
- `results/tables/E06_nstar.json` and `results/tables/E06_nstar_efficiency.json`;
- `results/tables/E16_quantum_uncertainty.json`;
- `results/tables/E08v2_independent_mc.json` and `results/tables/E08v3_multidraw.json`;
- `results/tables/E13v2_baw_allocation.json`.

The frozen protocols and post-run dispositions are in `docs/experiment_registry.md`, `docs/decisions.md`, and the three dated files under `docs/audits/`. These text records explain the design and corrections; the JSON artifacts remain the numerical source of truth.